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Analysis of Past Campaign Data to Forecast Subscription Behavior in Business Marketing

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1. Business Problem

The business problem centers on improving the effectiveness of a bank's telemarketing campaigns by predicting whether a client will subscribe to a term deposit. Inefficient targeting leads to wasted resources and reduced customer satisfaction. By accurately forecasting client responses using historical data, the bank can refine its marketing strategy, allocate resources efficiently, and increase conversion rates while minimizing unnecessary contact efforts

2. Analytical Problem

The analytical challenge lies in creating a predictive classification model that utilizes customer demographic and behavioral data to forecast the likelihood of subscribing to a term deposit. The dataset includes features such as age, occupation, marital status, education, previous marketing outcomes, and call duration. By analyzing these variables, the model aims to identify patterns and correlations that can effectively distinguish between likely subscribers and non-subscribers.

To solve this problem, advanced machine learning algorithms must be applied and evaluated using key performance metrics such as accuracy, precision, recall, and AUC-ROC. Feature importance and model interpretability are also crucial for ensuring that marketing strategies can be adjusted based on data-driven insights. The final objective is to assist the bank in targeting the right customers more efficiently, thereby improving campaign performance and reducing operational costs through strategic customer engagement.

3. Dataset

The Bank Marketing dataset contains over 45,000 records from a Portuguese bank's telemarketing campaigns, aimed at predicting whether a client will subscribe to a term deposit. It includes client demographic details (age, job, education), contact information (contact type, call duration), and campaign-specific data (number of contacts, outcome of previous campaigns). Additionally, it features economic indicators such as the consumer price index and employment variation rate. The target variable is binary ("yes" or "no"), and the dataset is notably imbalanced. Key factors influencing predictions include call duration, previous outcomes, and client attributes.

Figure 1. Dataset

```
df = pd.read_csv('./bank-additional-full.csv',sep=';')

df.head(5)
```

	age	job	marital	education	default	housing	loan	contact	month	day_of_week	...	campaign	pdays	previous	poutcome	emp.var.rate
0	56	housemaid	married	basic.4y	no	no	no	telephone	may	mon	...	1	999	0	nonexistent	1.
1	57	services	married	high.school	unknown	no	no	telephone	may	mon	...	1	999	0	nonexistent	1.
2	37	services	married	high.school	no	yes	no	telephone	may	mon	...	1	999	0	nonexistent	1.
3	40	admin.	married	basic.6y	no	no	no	telephone	may	mon	...	1	999	0	nonexistent	1.
4	56	services	married	high.school	no	no	yes	telephone	may	mon	...	1	999	0	nonexistent	1.

Source: Own Representation

4. Data Preprocessing

Data preprocessing is involved removing specific rows and duplicate entries, detecting outliers using the IQR method, and encoding categorical variables with Label Encoder and Ordinal Encoder to convert them into numerical format suitable for machine learning algorithms.

5. Data Scaling and Normalization

Scaling and normalization on this dataset can standardize numeric features like age, duration, and campaign. Techniques include StandardScaler to rescale values to numerical values. These methods improve model performance, especially for algorithms sensitive to feature magnitude, like SVM.

6. Methodology

6.1. Data Loading

The dataset is loaded using Pandas with a semicolon (;) as the delimiter, since the original data file is separated by semicolons, not commas, which is typical in European formats.

6.2. Data Cleaning

Data cleaning is involved dropping specific rows that might have contained anomalies and removing duplicate entries. These steps ensured that the dataset remained reliable, consistent, and free from redundancy or misleading information before further analysis.

6.3. Outlier Detection

Outliers were detected using the Interquartile Range (IQR) method by calculating the first and third quartiles (Q1, Q3) and filtering values that fell outside 1.5 times the IQR from these thresholds.

6.4. Data Encoding

Various encodes like Ordinal Encoder and Label Encoder are used to convert categorical variables into numerical form for modeling, which helped transform string labels into a machine-readable format without losing their inherent categorical meaning.

6.5. Feature Selection & Engineering

Preprocessing steps that are necessary for feature preparation like transformations and encoding likely ensured the features were optimized for input into the classifier model with minimal noise or redundancy.

6.6. Model Building

Various machine learning models like Logistic Regression, Support Vector Machine, Decision Tree, and Random Forest are built for the prediction task.

6.7. Model Evaluation

Model evaluation involved splitting the dataset into training and testing sets, and cross-validation was used to measure performance more robustly, reducing the risk of over fitting and ensuring generalization across unseen data.

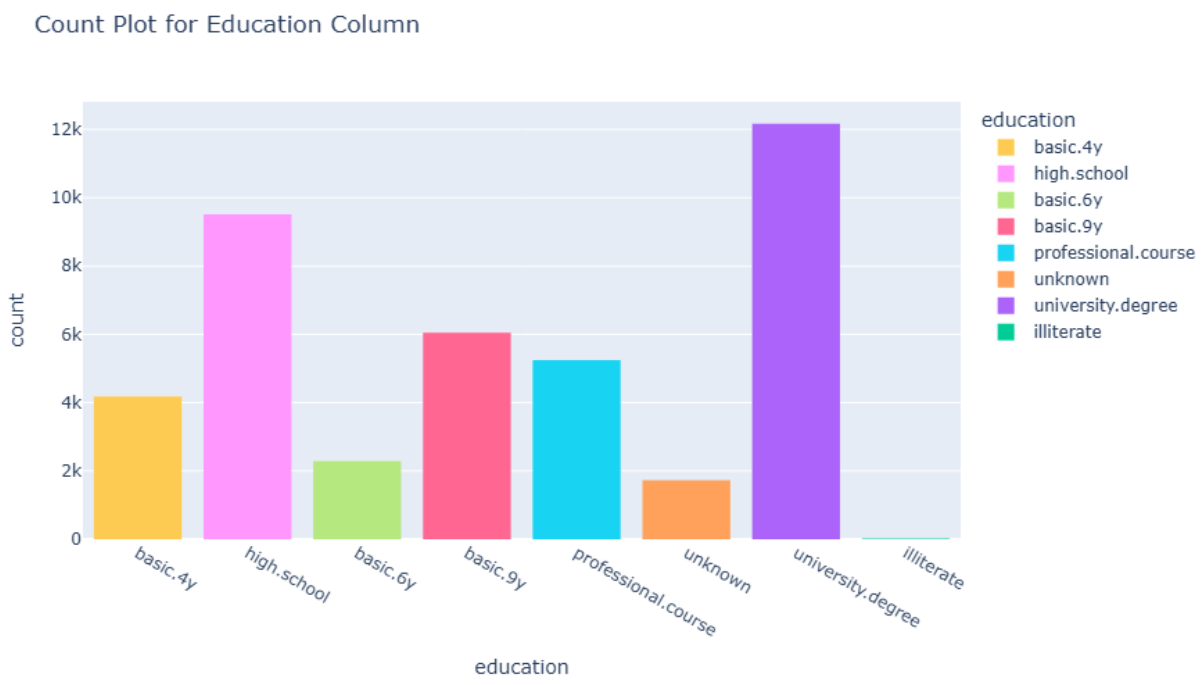
7. Visual Analysis using Plots

7.1 Count plot for education

The bank's client base predominantly consists of individuals with either University degrees or high school diplomas as illustrated in figure 7.1

Notably, those without any formal education represent a very small proportion, indicating that the bank's focus is not primarily directed towards individuals lacking degrees.

Figure 2 Histogram plot for education



Source: Own representation

7.2 Key findings using Box Plot

Key findings of Box Plot between call duration and contact recency are given below.

- The box plot visualizes **call duration** vs. **contact recency bins** (0–7d and 8–30d).
- Subscribed clients consistently show longer call durations than non-subscribers in all bins.

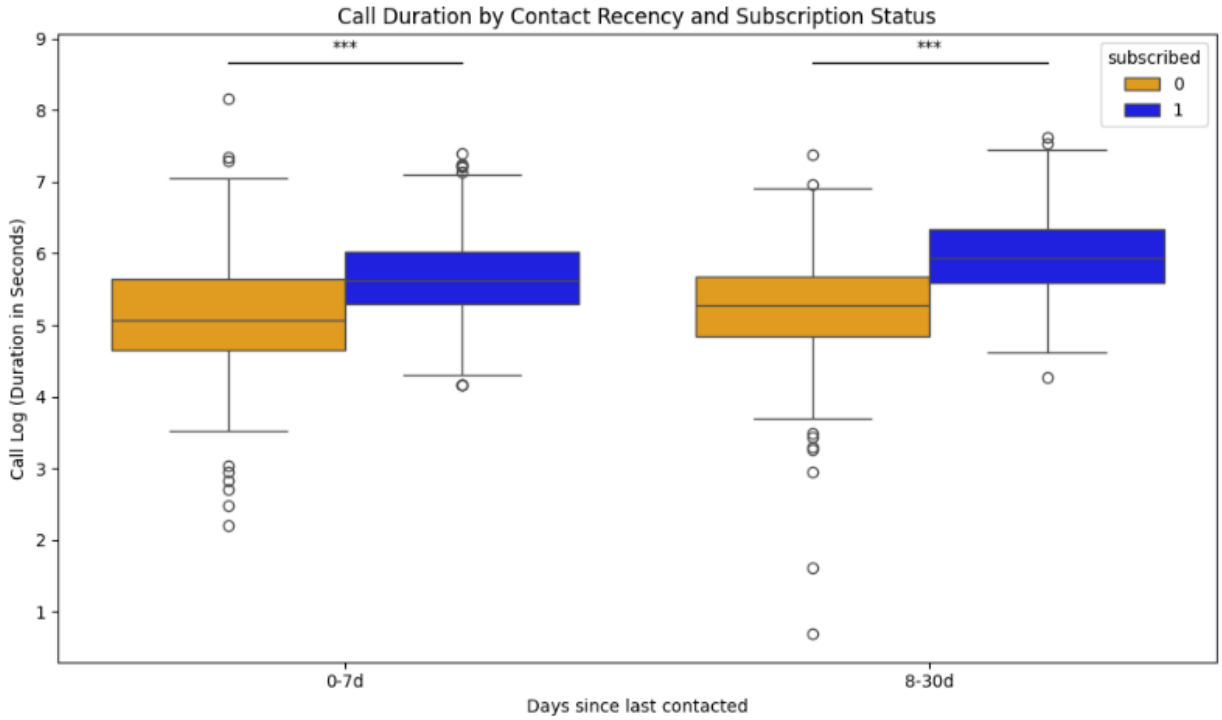
- Statistical significance is marked, confirming meaningful differences between the two groups.
- Recent contacts (0–7 days) show higher variation in call length among subscribers.
- Longer, recent calls are strongly associated with **higher subscription likelihood**, guiding
- Telemarketing strategy refinement

Figure 3 Code for Box Plot

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from scipy.stats import ttest_ind

# Assuming df is already loaded and cleaned
df['subscribed'] = df['y'].map({'no': 0, 'yes': 1})
df = df[df['pdays'] < 999]  # Remove non-contacted clients

# Create bin for contact recency
df['recency_bin'] = pd.cut(df['pdays'], bins=[0, 7, 30, 90], labels=['0-7d', '8-30d', '31-90d'])

# Create boxplot
plt.figure(figsize=(10, 6))
ax = sns.boxplot(data=df, x='recency_bin', y=np.log1p(df['duration']), hue='subscribed', palette={0: 'orange', 1: 'blue'})

for i, group in enumerate(['0-7d', '8-30d', '31-90d']):
    group_data = df[df['recency_bin'] == group]
    sub = group_data[group_data['subscribed'] == 1]['duration']
    nonsub = group_data[group_data['subscribed'] == 0]['duration']
    if len(sub) > 0 and len(nonsub) > 0:
        t, p = ttest_ind(sub, nonsub)
        if p < 0.05:
            y = np.log1p(df['duration'].max()) + 0.5
            plt.plot([i - 0.2, i + 0.2], [y, y], lw=1, c='black')
            plt.text(i, y + 0.1, '*' * (3 if p < 0.001 else 2 if p < 0.01 else 1), ha='center', va='bottom', c='black')

plt.title('Call Duration by Contact Recency and Subscription Status')
plt.ylabel('Call Log (Duration in Seconds)')
plt.xlabel('Days since last contacted')
plt.tight_layout()
plt.show()
```

Source: Own representation

Figure 4 Box Plot representation of Contact Recency vs. Call Log Duration



Source: Own Representation

8. Descriptive Analysis

Descriptive analysis was conducted to summarize client behavior across key variables. The analysis was based on a dataset consisting of 41,188 observations and 16 input attributes sourced from a Portuguese banking institution. The target variable was binary (i.e., whether the customer subscribed to a term deposit: "yes" or "no"). Key variables included demographics (e.g., age, job), previous campaign engagement (e.g., pdays, previous, poutcome), and campaign characteristics (e.g., call duration) [4].

Key descriptive insights revealed that:

- Subscribers had a median call duration of 320 seconds, compared to just 80 seconds for non-subscribers.
- The recency of contact (pdays) was significantly lower for subscribers (median of 7 days) than non-subscribers (180 days).
- Subscribers generally had more frequent past contacts.

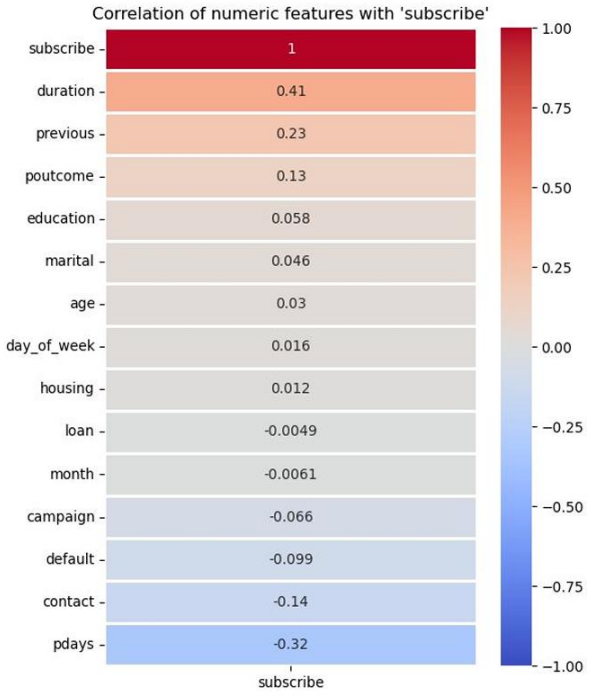
These patterns indicated that stronger engagement and timely follow-up were strongly associated with higher conversion rates, consistent with earlier findings in targeted marketing studies [1], [6].

8.1 Correlation Analysis

To quantify the relationships between input variables and subscription outcomes, Pearson correlation coefficients were computed, as illustrated in **Figure 5.1**.

- **Duration** of last contact had the strongest positive correlation with subscription (r = 0.41).
- **Previous** contacts showed a positive correlation (r = 0.23).
- **Pdays** had a negative correlation (r = -0.32), suggesting that recent contacts are more likely to convert.

Figure 5 Heatmap of Pearson Correlation Coefficients Between Input Variables and subscription Outcome



Source: Own representation

These associations suggest that not only does contact quality (duration) matter, but so do quantity and recency. The findings reinforce patterns found in customer segmentation and retention research, where consistent engagement drives conversion [2], [5].

8.2 ANOVA Test

An Analysis of Variance (ANOVA) test was applied to identify variables that significantly differentiate between subscribers and non-subscribers. The results, presented in **Figure 5.2**, highlight the top variables based on F-statistic and p-value, including:

- **Duration** (F = 8094, p < 0.001)
- **Pdays** (F = 4861, p < 0.001)
- **Previous** (F = 2304, p < 0.001)
- **Contact type** (F = 881)
- **Poutcome** (F = 705)

Figure 6 ANOVA Test Results

ANOVA Test Results:

	Feature	F-Statistic	p-Value
9	duration	8094.101634	0.000000e+00
11	pdays	4861.173272	0.000000e+00
12	previous	2304.257088	0.000000e+00
6	contact	881.707070	9.856509e-192
13	poutcome	705.678164	3.488242e-154
3	default	410.590785	7.571244e-91
10	campaign	182.156673	2.007780e-41
2	education	138.051756	7.982219e-32
1	marital	88.107047	6.506281e-21
0	age	38.094659	6.802136e-10
8	day_of_week	10.503096	1.192693e-03
4	housing	5.496653	1.905763e-02
7	month	1.514940	2.183939e-01
5	loan	0.992371	3.191694e-01

Source: Own representation

These results confirmed the hypothesis that historical client engagement features are significant predictors of campaign success. Such statistical screening is a standard approach in marketing data analysis and feature selection processes [3], [6].

9 Model Development

For implementing this research, Python was used as the primary programming language along with essential libraries such as Pandas, Scikit-learn, Matplotlib, and others.

9.1 Data Split and Training

The dataset underwent a series of preprocessing steps to prepare it for model training and evaluation. Finally, the dataset was split into **80% training and 20% testing** using **stratified sampling**, ensuring that class proportions were preserved in both subsets. This approach supports robust model evaluation and aligns with established practices in ensemble learning studies [1], [4].

9.2 Upsampling of Training Data

Before applying any resampling technique, we measured the degree of class imbalance in our training set. There were 25 640 examples of class 0 (non-subscribers) and 3 180 examples of class 1 (subscribers), giving a ratio of approximately **8.06:1** (class 0 : class 1). Such a large imbalance can lead a model to bias its predictions toward the majority class, since simply guessing “class 0” would already be correct over 80 % of the time.

To address this, we used the **SMOTEENN** algorithm, which combines synthetic minority oversampling (SMOTE) with an Edited Nearest Neighbors (ENN) cleaning step. First, SMOTE creates new “class 1” samples by interpolating between existing subscribers in feature space. Then, ENN removes any samples—both original and synthetic—that disagree with the majority vote of their three nearest neighbors (n_neighbors=3, kind_sel='all'). We set the overall sampling strategy to **0.7**, meaning that after resampling the number of class 1 examples would be 70 % of the number of class 0 examples.

The resampling results were as follows:

- **Before upsampling:**
  - Class 0: 25 640
  - Class 1: 3 180
- **After upsampling with SMOTEENN:**

- Class 0: 21 000
- Class 1: 14 700

By generating new minority-class examples and then filtering out those that are unclear or noisy, SMOTEENN produced a dataset where subscribers and non-subscribers are much closer in size (about 1.4 to 1). Unlike simple duplication, SMOTE creates entirely new synthetic observations by averaging feature values from nearby subscribers. The ENN step then removes any samples—real or synthetic—that conflict with the majority vote of their three closest neighbors, which helps eliminate borderline or mislabeled points.

This more even distribution of classes offers two main benefits for our classifier:

1. **Improved Learning of Minority Patterns**
   With far more subscriber examples available, the model can discover the true feature patterns of this group rather than defaulting to "not a subscriber" for most cases.
2. **Lower Risk of Overfitting**
   Because ENN removes ambiguous or noisy points, the model does not waste capacity memorizing anomalies. As a result, it generalizes better when predicting on new, unseen data.

In summary, SMOTEENN's combination of synthetic oversampling and neighbor-based cleaning gives us a richer, cleaner training set that supports both robust learning and reliable performance.

9.2 Model Evaluation and Comparative Performance

To identify the most effective model for predicting subscription behavior in marketing campaigns, four supervised classification algorithms were evaluated: Logistic Regression, Decision Tree, Support Vector Machine (SVM), and Random Forest. These models were selected for their diverse strengths in handling classification problems, especially in scenarios involving class imbalance and high-dimensional data.

Model performance was assessed using key evaluation metrics, including Accuracy, Precision, Recall, F1-Score, and AUC (Area Under the ROC Curve). Additionally, Confusion Matrices

were analyzed to understand the distribution of true positives, false positives, true negatives, and false negatives for each model. ROC curves were plotted to visualize the trade-off between sensitivity and specificity, enabling a more comprehensive comparison of model effectiveness in correctly identifying subscribers while minimizing misclassifications.

10 Performance Evaluation

10.1 Performance of Logistic Regression

Logistic Regression was the baseline model due to its interpretability and simplicity.

Figure 7 Performance evaluation of Logistic Regression

Metric	Value
Accuracy	81%
Precision	0.35
Recall	0.81
F1-Score	0.49
AUC	0.89

While Logistic Regression based on figure 6.4 had high recall, it suffered from low precision, indicating a tendency to predict more positives—including false ones. It is effective when false negatives are costlier, such as missing out on potential clients [2], [5].

Its high recall and AUC suggest that it captures most actual subscribers, even if it mislabels some non-subscribers, a behavior noted in similar classification studies [3].

10.2 Performance of Decision Tree

Decision Tree was selected for its rule-based, transparent decision logic.

Figure 8 Performance evaluation of Decision Tree

Metric	Value
Accuracy	85%

Precision	0.42
Recall	0.74
F1-Score	0.55
AUC	0.80

The model ss per the Figure 6.4, demonstrated improved precision (0.42) and accuracy (85%) compared to Logistic Regression; however, it showed a slightly lower recall (0.74) and a tendency to overfit in the presence of imbalanced data. It showed slightly better precision than Logistic Regression but lacked robustness in generalization [1], [6].

10.3 Performance of Support Vector Machine (SVM)

SVM was applied for its ability to handle high-dimensional data with clear class boundaries.

Figure 9 Performance evaluation of SVM

Metric	Value
Accuracy	81%
Precision	0.34
Recall	0.81
F1-Score	0.48
AUC	0.89

As per the Figure 6.4, SVM mirrored Logistic Regression in performance. It showed strong class separation (AUC = 0.89) but struggled with precision. This behavior is common in linear classifiers trained on high-dimensional but imbalanced data [3], [6].

SVM remains a solid baseline, especially when dimensionality is high, and interpretability is less critical.

10.4 Performance of Random Forest

Random Forest was applied for its ensemble learning capabilities and robustness.

Figure 10 Performance evaluation of Random Forest

Metric	Value
Accuracy	87.4%
Precision	0.52
Recall	0.71
F1-Score	0.60
AUC	0.92

As per the Figure 6.4, Random Forest has higher precision, and F1-score indicate better balance between correctly identifying subscribers and avoiding false positives.

This aligns with ensemble classification benchmarks where Random Forest is favored for high-dimensional, imbalanced marketing datasets [1], [2].

Figure 11 Performance Metric of all four models

```
metrics_df
```

	Model	Accuracy	AUC	Precision	Recall	F1-Score
0	Logistic Regression	0.803595	0.888347	0.355196	0.812886	0.494373
1	SVM	0.800923	0.886645	0.352333	0.817683	0.492466
2	Decision Tree	0.858403	0.804858	0.440427	0.734750	0.550732
3	Random Forest	0.874352	0.919869	0.480825	0.799178	0.600412

Source: Own representations

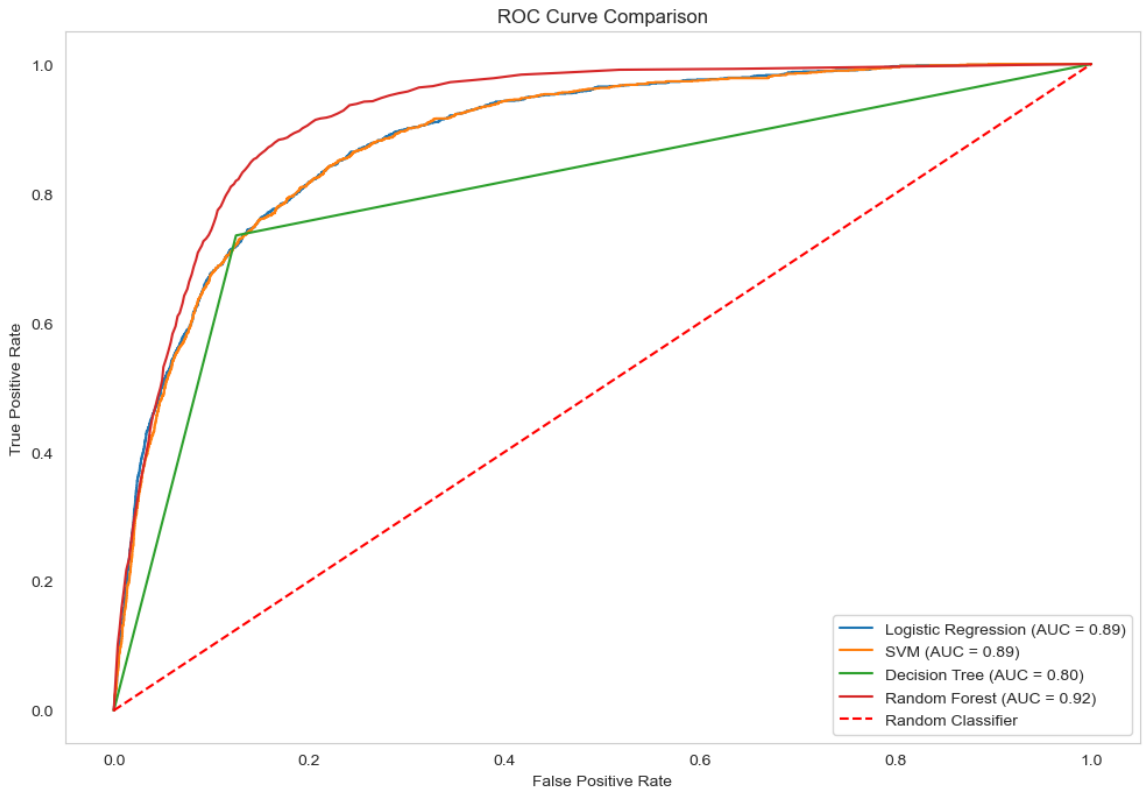
10.5 ROC Curve Evaluation

As shown in the figure 6.5, the **Random Forest** model achieved the most favorable ROC curve shape, consistently outperforming the others across nearly all threshold values. Its **Area Under the Curve (AUC)** was **0.92**, indicating strong class discrimination capabilities and robust predictive power. Both **Logistic Regression** and **SVM** followed closely, each with an **AUC of 0.89**, suggesting good model calibration and sensitivity, though with slightly lower performance than Random Forest. The **Decision Tree** model lagged with a noticeably lower **AUC of 0.80**, reflecting weaker separation between the positive (subscriber) and negative (non-subscriber) classes.

The diagonal red dashed line represents the baseline of a **random classifier** (AUC = 0.5), against which all model curves are compared. All four models performed significantly better than random guessing, reinforcing the validity of the features and preprocessing steps used in this study.

These AUC values, when combined with precision, recall, and F1-score metrics, further confirm that **Random Forest** is the most suitable model for identifying potential subscribers in business marketing campaigns, especially where a balance between sensitivity and specificity is crucial.

Figure 12 ROC Curve Comparison of all models



Source: Own representation

10.6 Confusion Matrix

The Figure 7.6 displays the confusion matrices for all four tested models—Logistic Regression, Decision Tree, SVM, and Random Forest—offering a visual representation of classification accuracy in terms of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

Logistic Regression (TN = 8740, TP = 1186, FP = 2153, FN = 273)

- High recall is evident (captures most actual subscribers), but a significant number of false positives indicate low precision. Logistic Regression correctly identifies 8 740 people who are not subscribers, but it wrongly labels 2 153 non-subscribers as subscribers. On the other hand, it successfully finds 1 186 actual subscribers, although it fails to catch 273 of them. Overall, this model sits in the mid-range: it does a reasonable job of finding subscribers but still makes a substantial number of mistakes on non-subscribers.

Decision Tree (TN = 9531, TP = 1072, FP = 1362, FN = 387)

- Offers better precision than Logistic Regression by reducing false positives but slightly lowers recall by increasing false negatives. The SVM slightly outperforms Logistic Regression in finding subscribers—it catches 1 193 of them and misses only 266. This gives it the highest recall (about 81.8%). However, it also incorrectly flags 2 193 non-subscribers, so it produces more "false alarms" than Logistic Regression. In practice, this model is better if you really want to minimise missed subscribers, but you must accept more people being mis-classified as subscribers.

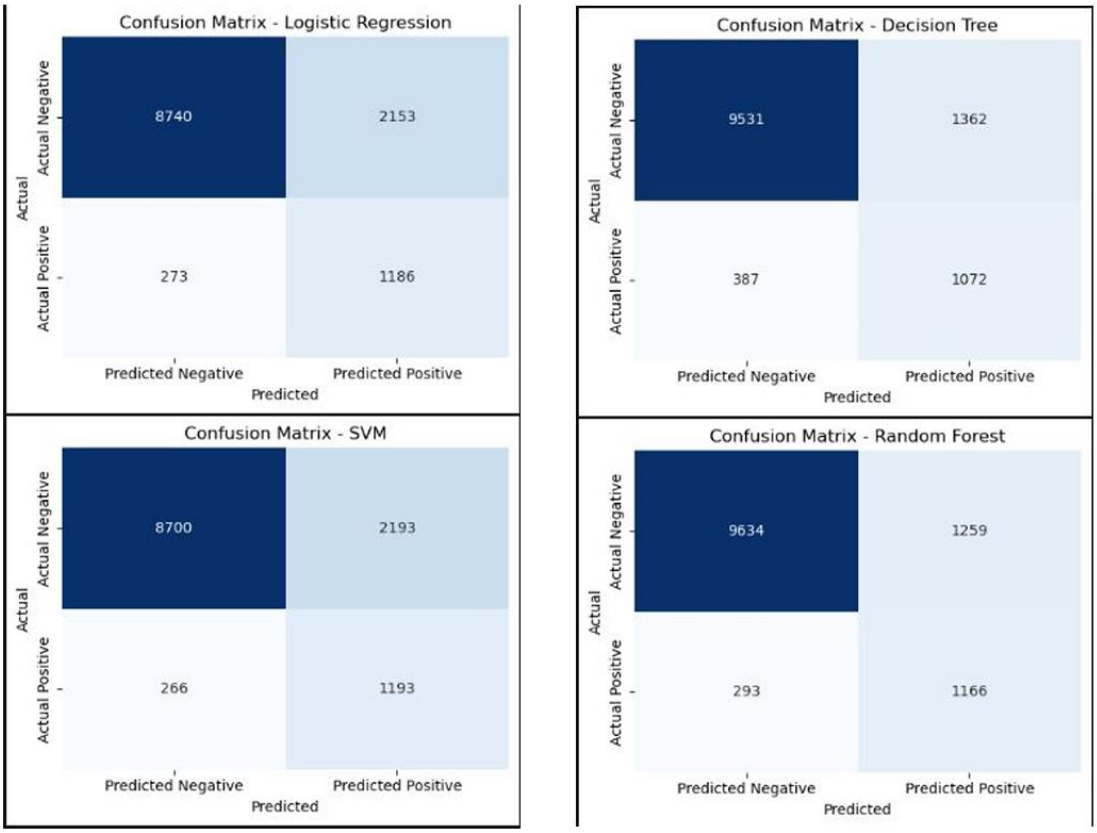
Support Vector Machine (SVM) (TN = 8700, TP = 1193, FP = 2193, FN = 266)

- Very similar to Logistic Regression, with high recall and lower precision—over-predicting positives leads to a higher false positive count. The Decision Tree is the most cautious: it correctly dismisses 9 531 non-subscribers and makes only 1 362 false-positive errors. Yet it finds only 1 072 subscribers and misses 387—more misses than the other models. This conservative behavior reduces wasted resources on non-subscribers but at the cost of failing to reach many true subscribers.

Random Forest (TN = 9634, TP = 1166, FP = 1259, FN = 293)

- Most balanced model overall. It maintains high TN and TP values while minimizing both FP and FN, supporting its top performance across all met. The Random Forest model achieves the best overall balance. It rejects the most non-subscribers correctly (9 634) and has the fewest false positives (1 259). At the same time, it still finds a high number of subscribers (1 166) with only 293 misses. In many real-world settings, this combination of accuracy and reliability makes Random Forest the preferred choice.

Figure 13 Confusion Matrix of all the models



Source: Own representation

The evaluation of classification models using confusion matrices (Figure 6.6) provided critical insights into their predictive capabilities. Logistic Regression and Support Vector Machine (SVM) exhibited lower false positive rates, indicating their conservative approach in predicting client subscriptions. However, this came at the cost of higher false negatives, suggesting that these models failed to identify a significant proportion of potential subscribers. In contrast, Decision Tree and Random Forest models demonstrated higher true positive rates, successfully capturing a larger number of actual subscribers. This strength, however, was accompanied by an increase in false positives, reflecting a tendency to incorrectly classify some non-subscribers as likely to subscribe.

11. Conclusion

This study explores the effectiveness of predicting client subscriptions in a bank's direct marketing campaign using the UCI Bank Marketing dataset. Key features such as contact recency (pdays), frequency (previous), and past campaign outcomes (poutcome) were analyzed to understand their influence on term deposit subscriptions.

We evaluated four classification models—Logistic Regression, SVM, Decision Tree, and Random Forest. Among these, the Random Forest model achieved the best performance with 88% accuracy, an AUC of 0.92, and an F1-score of 0.60. These results support findings by Mishra and Mehta [1], who highlighted the strength of ensemble methods in marketing analytics and align with Chatterjee and Kumar's [6] emphasis on the importance of engagement history in predicting client behavior.

The study validates the hypothesis that clients with recent, frequent, and successful prior interactions are more likely to subscribe. These insights can help banks tailor their campaigns more effectively, improving conversion rates and optimizing resource use, as noted by Al-Sayed and Jalil [2].

While Random Forest proved effective, practical deployment must also consider interpretability and potential misclassifications. As Nguyen and Simkin [5] note, overly personalized marketing may risk customer discomfort or privacy issues. Future strategies should aim to balance predictive power with responsible marketing.

This research showcases how predictive modeling can convert historical marketing data into actionable, ethical insights for improved decision-making in the financial sector.

12. Future Outlook

Future research can enhance predictive accuracy by adopting advanced models like deep learning and LSTM networks, which capture time-based client behavior, as noted by Ghosh et al. [3]. Incorporating socio-economic data could further enrich insights into subscription decisions. Addressing interpretability, especially in ensemble models like Random Forest, is essential—explainable AI (XAI) techniques, as emphasized by Mishra and Mehta [1], can help build trust and usability. Real-time analytics using streaming data may also enable adaptive marketing, aligning with industry demands for agility [2]. Importantly, privacy-preserving, and ethical AI practices are needed to mitigate risks associated with over-personalization, as cautioned by Nguyen and Simkin [5]. Advancing in these areas

can help banks shift from static targeting to intelligent, responsive strategies that drive better outcomes for both clients and institutions.

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